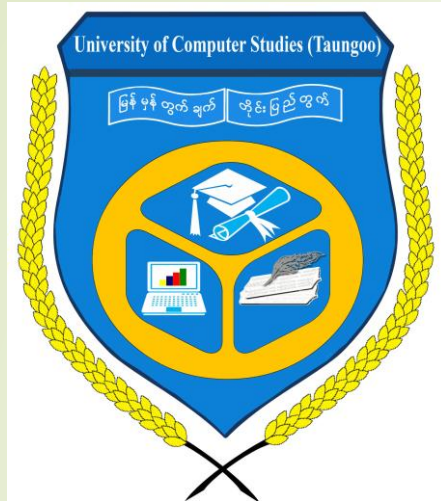




Introduction To Python

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Contents

▀ Functions

Function

Creating a function

- A function is defined using the **def** keyword.

Example:

```
def funname():  
    print("This is a function creation")
```

- To call a function, the function name is used and followed by parenthesis.

```
funname()
```

Function

Parameters and Arguments

- A parameter is the variable listed inside the parentheses in the function definition.
- An argument is the value that is sent to the function when it is called.

Example:

```
def funame(firstname, lastname):  
    print(firstname + " " + lastname)
```

```
funame("Song", "JongKi")    // function call
```

Function

Arbitrary Arguments (* args)

- If you do not know how many **arguments** that will be passed into your function, add ***** before the parameter name in the function definition.
- The function will receive a **tuple of arguments**, and can access the item accordingly.

Example:

```
def marketss(*fruits):  
    print("My favourite fruit is "+fruits[1])
```

```
marketss("Apple", "Orange", "Grape")    // function call
```

Function

Keyword Arguments

- You can also send arguments with **the key=value** syntax.
- It does not matter for the order of arguments.

Example:

```
def marketss(fruit1, fruit2, fruit3):  
    print("My favourite fruit is "+fruit2)
```

```
marketss(fruit1="Apple", fruit3="Grape", fruit2= "Orange")    // function call
```

Function

Arbitrary Keyword Arguments

- If you do not know how many **keyword arguments** that will be passed into your function, add ****** before the parameter name in the function definition.
- The function will receive a **dictionary of arguments**, and can access the item accordingly.

Example:

```
def marketss(**fruits):  
    print("My favourite fruit is "+fruits["fruit2"])
```

```
marketss(fruit1="Apple", fruit2="Orange", fruit3="Grape")    // function call
```

Function

Default parameter value

- If you call the function without argument, it uses the default value.

Example:

```
def marketss(fruit = "Orange"):  
    print("My favourite fruit is "+fruit)
```

marketss("Apple")

=> My favourite fruit is Apple

marketss("Grape")

=> My favourite fruit is Grape

marketss()

=> My favourite fruit is Orange

Function

Passing a List as an Argument

- You can send any data types of argument to a function(string, number, list,dictionary).
- If you send a list as an argument, it will still be a list when it reaches the function.

Example:

```
def marketss(fru):
```

```
    for x in fru:
```

```
        print(x)
```

```
fruits=["Apple", "Orange", "Grape"]
```

```
marketss(fruits)
```

Function

Return Values

- To let a function return a value, use the return statement.

Example:

```
def multifun(num):  
    return 2*num
```

```
x=multifun(5)  
print(x)           => 10  
print(multifun(10)) => 20
```

The pass Statement

- A function definition with no content, put in the pass statement to avoid getting an error.

Example:

```
def multifun( ):  
    pass
```

Function

Recursion

➤ It means a defined function can call itself.

Example:

```
def tri_recursion(k):  
    if(k>0):  
        result = k+tri_recursion(k-1)  
        print(result)  
    else:  
        result = 0  
    return result
```

```
print("\\nRecursion Result is ")  
tri_recursion(6)
```

Function

Lambda

- A lambda function is a small anonymous function.
- A lambda function can take any number of arguments, but can **only have one expression**.

Syntax

lambda arguments : **expression**

Example:

```
x= lambda a : a+10
```

```
print(x(5))
```

Example:

```
x= lambda a, b : a*b
```

```
print(x(5,6))
```

Example:

```
def myfun(n):
```

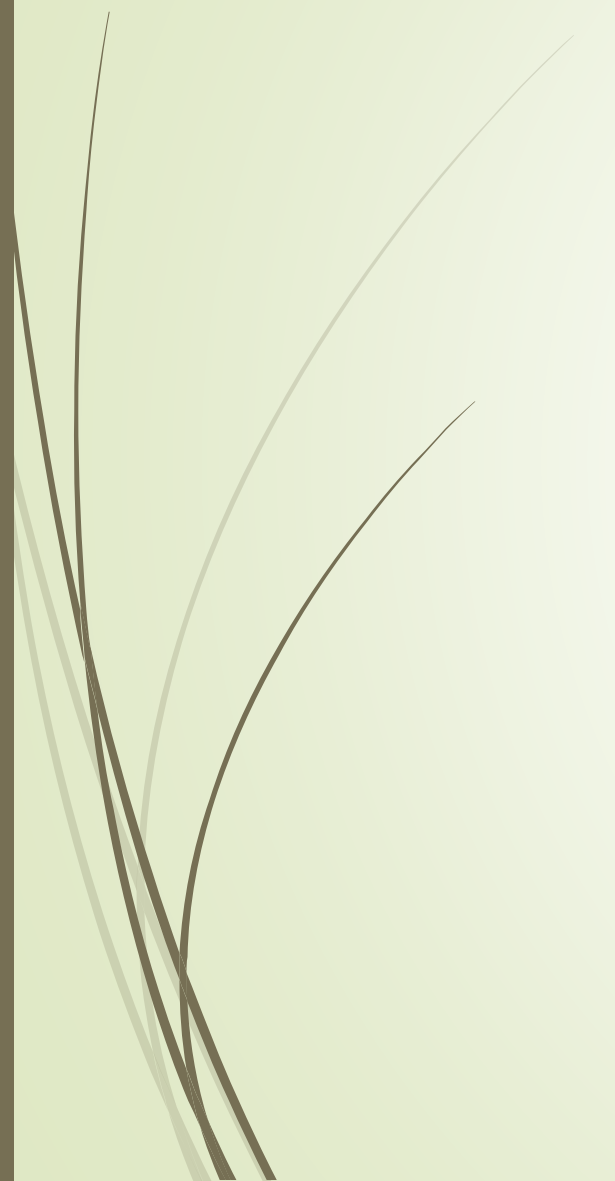
```
    return lambda a: a*n
```

```
mydoubler = myfun(2)
```

```
print(mydoubler(11))
```

Exercises

- <https://pynative.com/python-operators-and-expression-quiz/>
- <https://pynative.com/python-variables-and-data-types-quiz/>
- <https://pynative.com/basic-python-quiz-for-beginners/>
- <https://pynative.com/python/quizzes/>
- <https://pynative.com/python-basic-exercise-for-beginners/>
- https://www.w3schools.com/python/python_functions.asp



Thank You